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UNIZIK FRESHER EDITION | 2026

PHY 101: General Physics I - Mechanics, Matter and Waves

Build the physical intuition and problem-solving habits that General Physics rewards

FACULTY PRIMER

Faculty Primer

Faculty of Physical Sciences
Physics & Industrial Physics | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

This pack treats General Physics as a small number of powerful ideas applied repeatedly. You get a structured method for solving mechanics problems, a clear treatment of simple harmonic motion and waves, and a formula sheet with the conditions under which each formula is valid.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Solve kinematics and dynamics problems using a consistent sign convention and free-body diagrams.
- Apply conservation of energy and momentum to collisions and connected bodies.
- Explain simple harmonic motion and calculate period, frequency and energy.
- Describe wave properties including superposition, interference and standing waves.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Problem-solving protocol

Physics marks are lost in the setup, not in the algebra. Use the same five steps every time until they are automatic.

1. Draw a diagram and label known and unknown quantities with units.
2. Choose a coordinate system and state which direction is positive.
3. Write the relevant principle - Newton's second law, energy conservation, or momentum conservation.
4. Substitute symbols first, then numbers. This makes errors easier to spot and earns method marks.
5. Check magnitude, units and whether the direction of the answer is physically sensible.

Free-body diagrams are not optional

If two bodies are in contact or connected by a string, draw each body separately. Most 'impossible' pulley problems become simple once the tensions and weights are separated.

Mechanics essentials

Kinematics describes motion; dynamics explains it. Keep them distinct in your answers.

- Equations of motion apply only when acceleration is constant.
- Newton's second law: net force equals mass times acceleration. Net means the vector sum.
- Friction: static friction can take any value up to the limiting value; kinetic friction is roughly constant.
- Work done equals force times displacement in the direction of the force.
- Conservation of mechanical energy applies only when no non-conservative forces do work.
- Momentum is conserved in collisions; kinetic energy is conserved only in elastic collisions.

Oscillations and waves

Simple harmonic motion appears in springs, pendulums, molecules and circuits. Learn the defining relationship and the rest follows.

- Defining condition: acceleration is proportional to displacement and directed toward equilibrium.
- Period of a mass-spring system: T = 2 pi times the square root of m/k.
- Period of a simple pendulum for small angles: T = 2 pi times the square root of L/g.
- Total energy in SHM is proportional to the square of the amplitude.
- Wave speed equals frequency times wavelength.
- Superposition explains interference, beats and standing waves; nodes are points of zero displacement.

Angle in radians

The small-angle approximation sin(theta) is approximately theta only when theta is in radians. Using degrees silently corrupts the pendulum formula.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. A body starts from rest and accelerates at 2 m/s^2 for 5 s. Find its displacement.

Answer: s = ut + 0.5at^2 = 0 + 0.5 x 2 x 25 = 25 m.

2. State the condition for mechanical energy to be conserved.

Answer: Only conservative forces do work; no friction, air resistance or other non-conservative force removes energy from the system.

3. A spring of constant 200 N/m is stretched 0.1 m. Find the stored energy.

Answer: E = 0.5 k x^2 = 0.5 x 200 x 0.01 = 1 J.

4. A wave has frequency 50 Hz and wavelength 4 m. Find its speed.

Answer: v = f x wavelength = 50 x 4 = 200 m/s.

5. Why do we draw separate free-body diagrams for connected bodies?

Answer: Each body has its own forces and acceleration; separating them prevents incorrectly adding tensions and weights to the same object.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write the four constant-acceleration equations and state when they apply.
- [ ] Solve ten problems using free-body diagrams, one per problem.
- [ ] Derive the period of a mass-spring system in your own steps.
- [ ] Draw a standing wave and label nodes and antinodes.
- [ ] Build a one-page formula sheet with validity conditions, not just formulas.
- [ ] Do one timed past-question paper and mark every lost mark by cause.
- [ ] Explain one mechanics topic aloud to a coursemate without notes.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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